

Nikhil Rout

📍 Hyderabad, India ✉ nikhilrout97@gmail.com ☎ +91 6303583185 in nikhil-rout
🐙 NikhilRout 🔗 NikhilRout.github.io

Education

Vellore Institute of Technology, Chennai Campus	<i>2022 – 2026</i>
<i>Bachelor of Technology, Electronics and Communication Engineering</i>	<i>CGPA: 9.11/10</i>

Research Experience

Research Intern 🐙	<i>Remote</i>
Vortex GPGPU, ORCAS Lab, UCLA 🔗	<i>May 2025 – Present</i>

- Developed a Tensor Core Unit (TCU) extension for the open-source RISC-V extended Vortex GPGPU, enabling efficient WMMA operations for deep learning workloads
- Architected and implemented a high-performance configurable mixed-precision Fused Dot Product (FEDP) unit as part of the TCU in SystemVerilog
- Extended custom float conversions, SGEMM test, and cycle-level simulator implementation for FP8/BF8 support in C++
- Introduced a novel fused pipeline strategy for supporting Integer arithmetic within the floating-point datapath, maximizing component reuse while incurring minimal overhead
- Performed microarchitecture design space explorations, evaluating 3:2 vs 4:2 compressors in Carry-Save Adders, and format-wise Dedicated vs Shared multiplier schemes
- Reduced dynamic power consumption through clock/enable gating across FEDP pipeline stages during operation in 2:4 structured sparsity mode
- Contributed to TCU specific CI/CD and Verilator-based unittesting verification framework
- Achieved 362.2 MHz operation with 4-cycle latency on the Xilinx U55C FPGA, delivering 1.448 GFLOPS single-cycle throughput— $4.3\times$ the open-source baseline Berkeley HardFloat

Summer Research Intern 🐙	<i>Remote</i>
Centre for Nanoelectronics and VLSI Design, VIT Chennai 🔗	<i>May 2024 – July 2024</i>

- Implemented pipelined radix-2 FFT modules using both DIT and DIF approaches
- Developed Number Theoretic Transform (NTT) modules for efficient polynomial multiplication in lattice-based post-quantum cryptography schemes on FPGAs

Research Publications

-
- **A Configurable Mixed-Precision Fused Dot Product Unit for GPGPU Tensor Computation**
Presentation at Vortex Workshop and Tutorials, IEEE/ACM MICRO 2025 [Preprint](#) 🔗 🐙
 - **Optimizing RGB to Grayscale, Gaussian Blur and Sobel-Filter operations on FPGAs for reduced dynamic power consumption**
Presented at 3rd IEEE conference on AIIoT 2024 [IEEE Xplore](#) 🔗 🐙

Projects

Carry-Save FAN modification (SIGMA DNN Accelerator)

- Optimized Forward-Adder-Network (FAN) critical path by extending a Carry-Save-Adder tree structure for multi-vector reduction in sparse and irregular GEMM workloads
- Systolic Array Processing Element integration and synthesis PPA analysis in progress

Flash-Attention CUDA MegaKernel Inference Optimization

- Implemented CUDA kernels for Flash-Attention inference with baseline and Tensor Core variants, leveraging shared memory tiling and online softmax to reduce HBM bottleneck
- Profiled kernel performance on NVIDIA T4 GPU, demonstrating speedup from operator fusion and memory hierarchy optimization

Skills and Competencies

Languages: Verilog/SystemVerilog, C/C++, CUDA, Python, TCL, MATLAB

Tools: Quartus Prime, Xilinx Vivado/Vitis, Synopsys VCS/DC, CMake, Git, LaTeX

Relevant Coursework: Digital Systems Design, VLSI System Design, Computer Architecture, FPGA-based System Design, Heterogeneous Computing Systems, Deep Learning

References

Dr. Blaise-Pascal Tine

UCLA

ORCAS Lab, Assistant Professor

✉ blaisetine@cs.ucla.edu

Dr. Jean Jenifer Nesam J

VIT Chennai

Assistant Professor

✉ jeanjenifernesam.j@vit.ac.in

Dr. Umadevi S

VIT Chennai

CNVD, Associate Professor

✉ umadevi.s@vit.ac.in